

FeCode_updated2025_20260226

2026-02-27

Things that need to be changed

```
Date_Run = "20260226"
plates<- c("Plate4", "Plate5", "Plate6")
Std_plates<- c("STD")
Run_by = "Zoe Read"#Instrument user
Script_run_by = "Zoe Read"#Code user
Project = "COMPASS_Synoptic_CB"
Experiment="2025"
Run_notes="The Fe2 diff std curve R squared is 0.9844, which is slightly below the 0.985 cutoff but I t

#Stds that should be excluded
stds_to_remove<-c("")
```

File paths

```
## [1] "Tidy Data/20260226_COMPASS_Synoptic_CB_Fe_STD.csv"

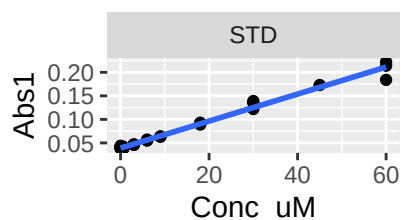
## [1] "Tidy Data/20260226_COMPASS_SynopticCB_Fe_Plate4.csv"
## [2] "Tidy Data/20260226_COMPASS_SynopticCB_Fe_Plate5.csv"
## [3] "Tidy Data/20260226_COMPASS_SynopticCB_Fe_Plate6.csv"
```

Plot standards

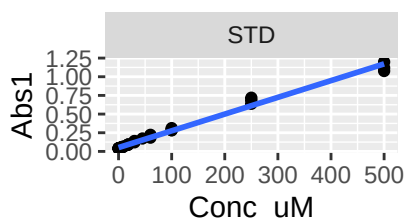
Table 1: Slopes

Date	Project	Curve	R2	Slope	Intercept	Top_STD
20260226	COMPASS_Synoptic_CFE_2_low		0.9862534	0.0028698	0.0389987	60
20260226	COMPASS_Synoptic_CFE_2_high		0.9901785	0.0022418	0.0514061	500
20260226	COMPASS_Synoptic_CFE_Tot_low		0.9926499	0.0018337	0.0401318	60
20260226	COMPASS_Synoptic_CFE_Tot_high		0.9928402	0.0018349	0.0401207	100
20260226	COMPASS_Synoptic_CFE_2_abs2		0.9967341	0.0043101	0.0417460	100
20260226	COMPASS_Synoptic_CFE_3_abs1		0.5084367	0.0000109	0.0396805	500
20260226	COMPASS_Synoptic_CFE_2_diff		0.9844930	0.0016862	0.0005130	100

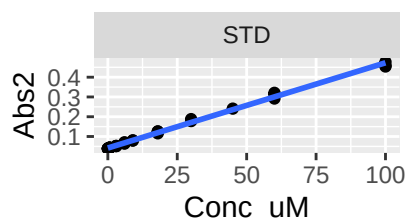
Low Fe(II) Absorbance



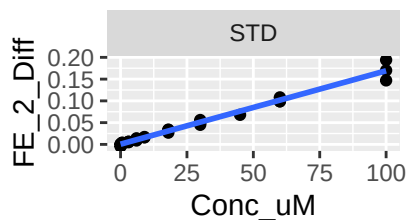
High Fe(II) Absorbance



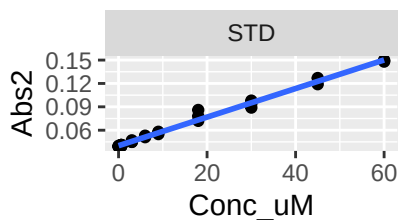
Fe(II) Absorbance 2



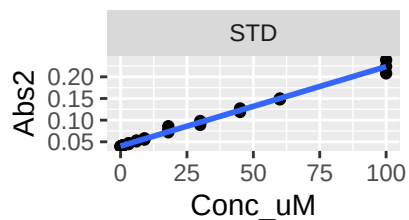
Fe(II) Absorbance D



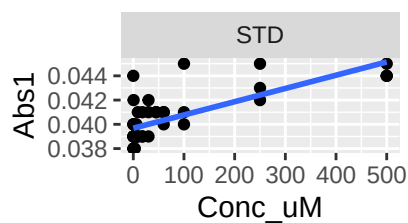
Low Fe(III) Absorbance



High Fe(III) Absorbance

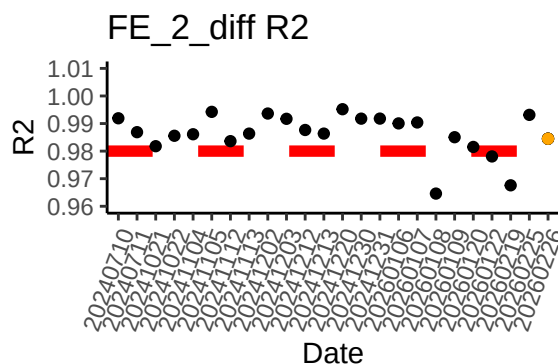
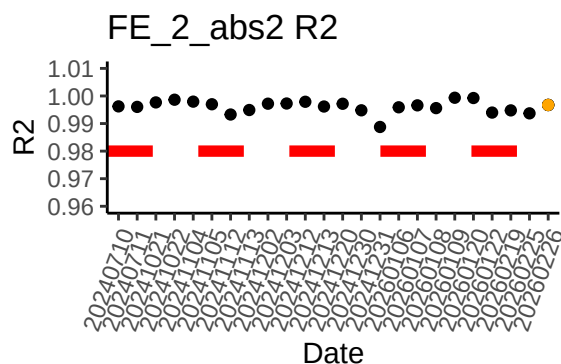
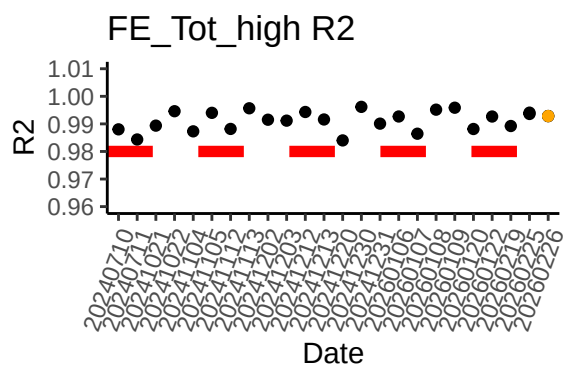
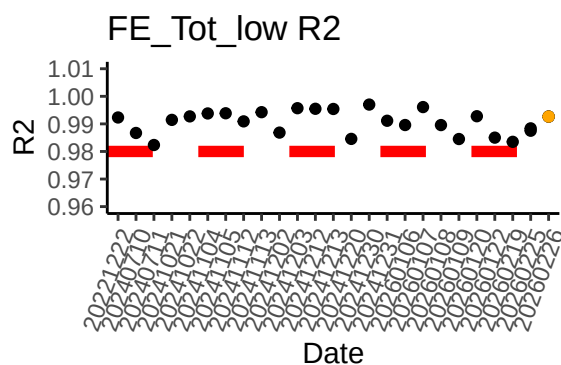
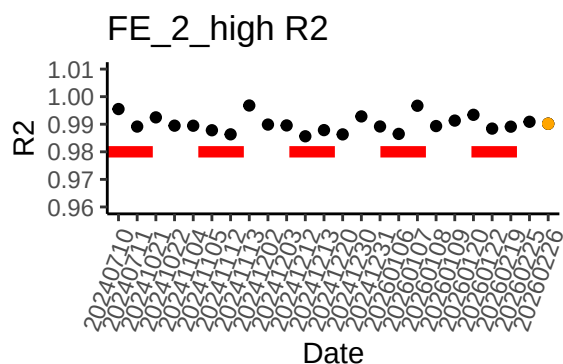
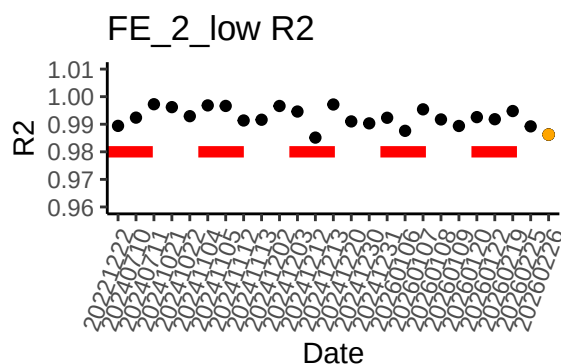


Fe(III) Absorbance 1 Curve



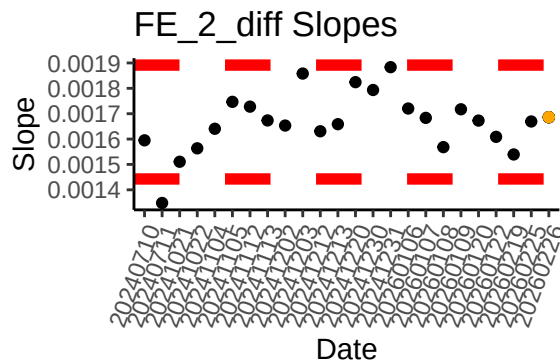
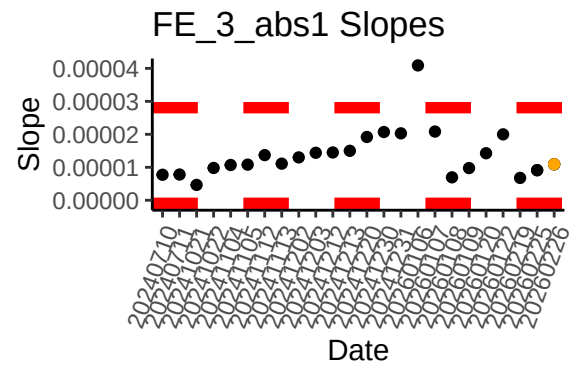
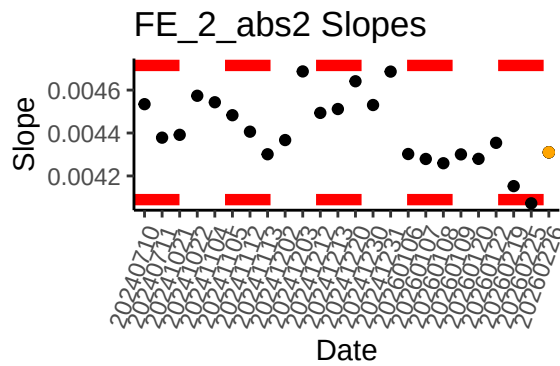
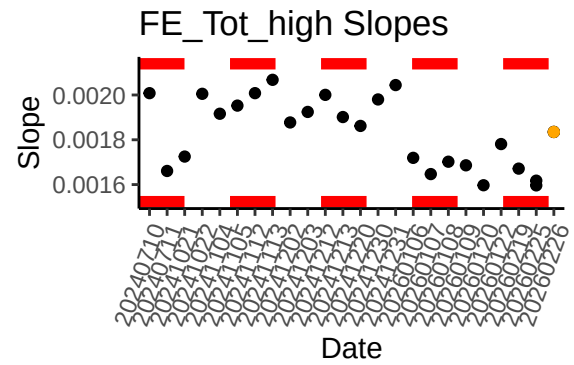
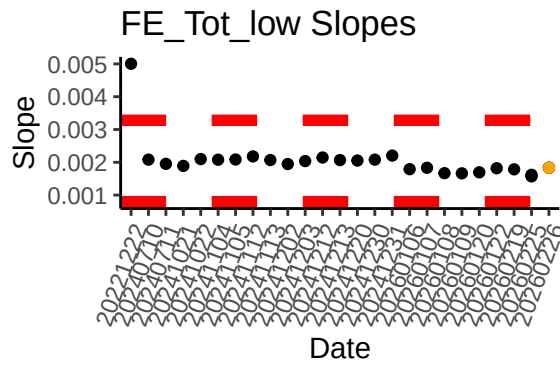
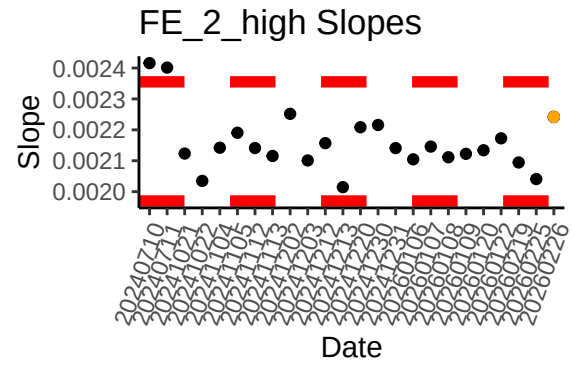
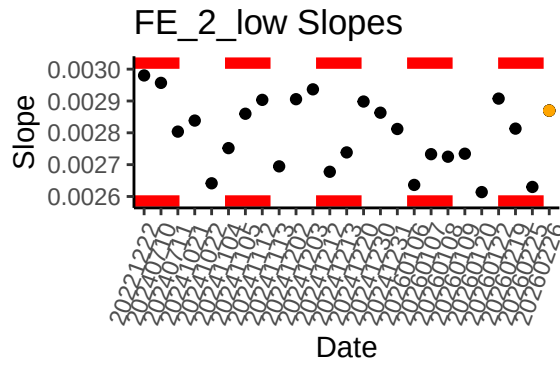
Checking STD Data against QAQC file

R2



```
## [1] "FE_2_low: Std Curve r2 GOOD"
## [1] "FE_2_high: Std Curve r2 GOOD"
## [1] "FE_Tot_low: Std Curve r2 GOOD"
## [1] "FE_Tot_high: Std Curve r2 GOOD"
## [1] "FE_2_abs2: Std Curve r2 GOOD"
## [1] "FE_3_abs1: R2 will always be low"
## [1] "FE_2_diff: Std Curve r2 is below cutoff! - REASSESS"
```

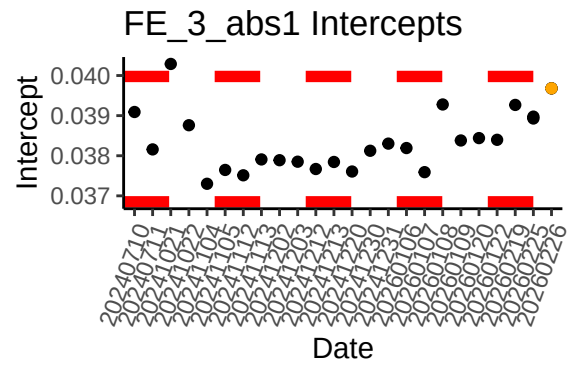
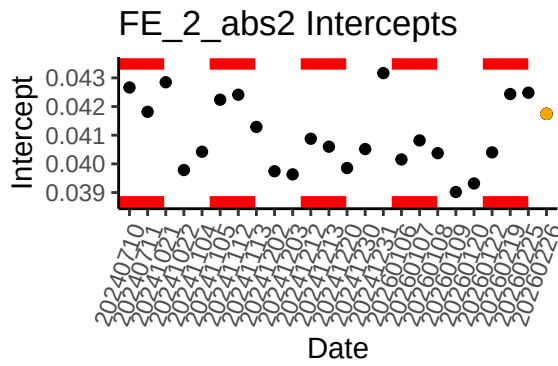
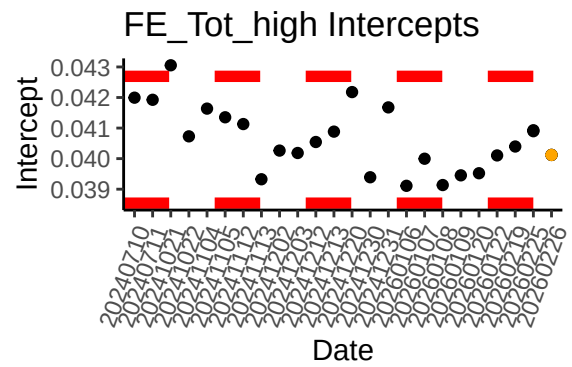
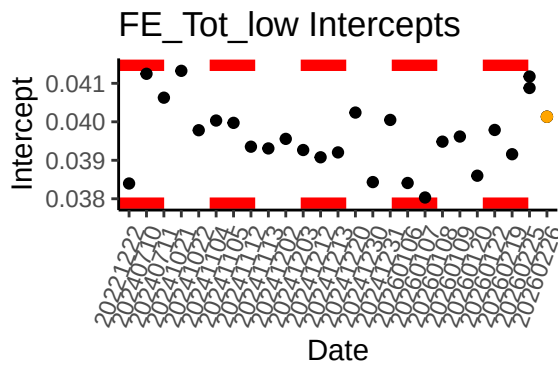
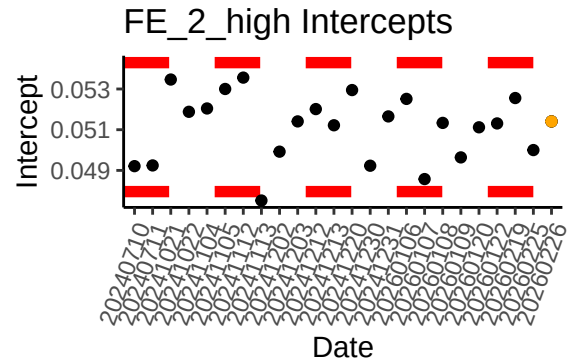
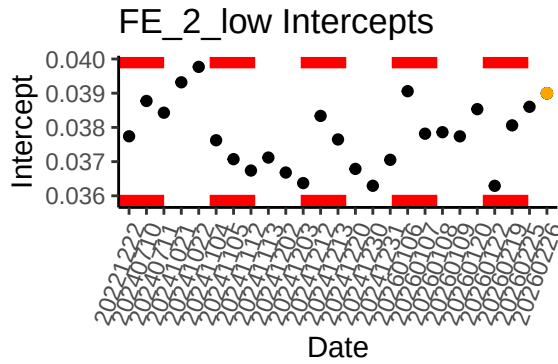
Slopes

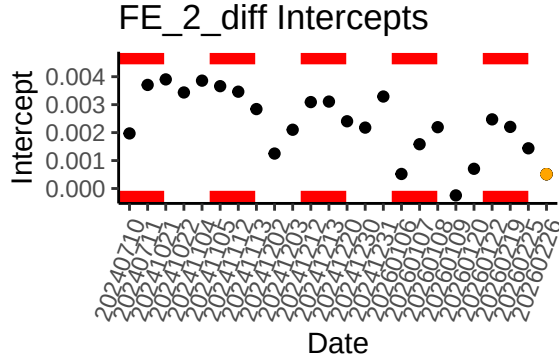


```
## [1] "FE_2_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_high:Std curve slope is with 2 sd of previous slopes"
```

```
## [1] "FE_Tot_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_Tot_high:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_abs2:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_3_abs1:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_diff:Std curve slope is with 2 sd of previous slopes"
```

Intercepts



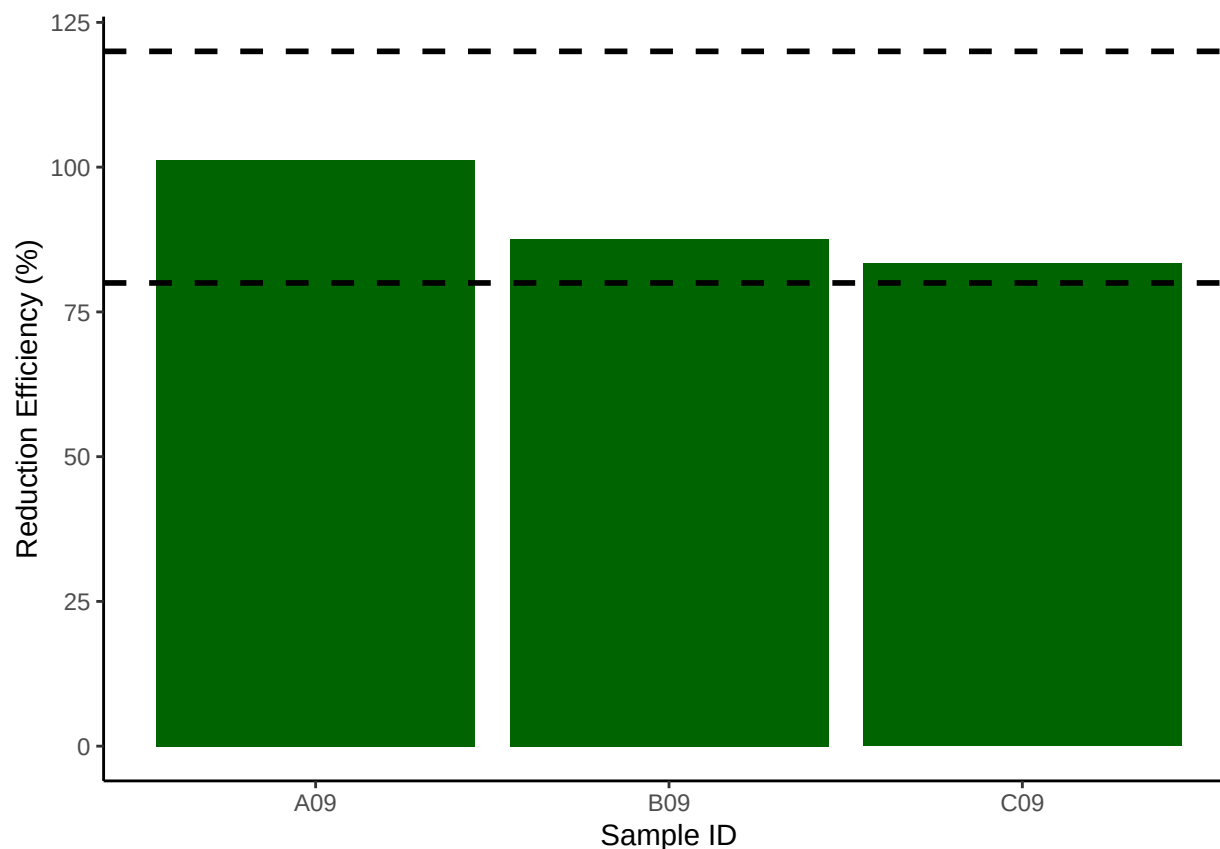


```
## [1] "FE_2_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_abs2:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_3_abs1:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_diff:Std curve intercept is with 2 sd of previous intercepts"
```

Calculate Reduction Efficiency

Table 2: Reduction Efficiency

ID	FE2	FE3	Eff	Eff1	Eff_flag
A09	0.184	0.149	80.97826	101.22283	OK
B09	0.214	0.150	70.09346	87.61682	OK
C09	0.222	0.148	66.66667	83.33333	OK



```
## [1] ">60% of Reduction Efficiencies are within range"
```

Method Minimum Detection Limit (MDL) Calculation

I use the lowest standard to calculate MDL How to calculate MDL https://www.epa.gov/sites/default/files/2016-12/documents/mdl-procedure_rev2_12-13-2016.pdf Look at Table 1 on page 5 The Student's t-value used to calculate the method detection limit (MDL) is the one appropriate for a 99% confidence level and a standard deviation estimate with n-1 degrees of freedom

```
## [1] 0.002415156
```

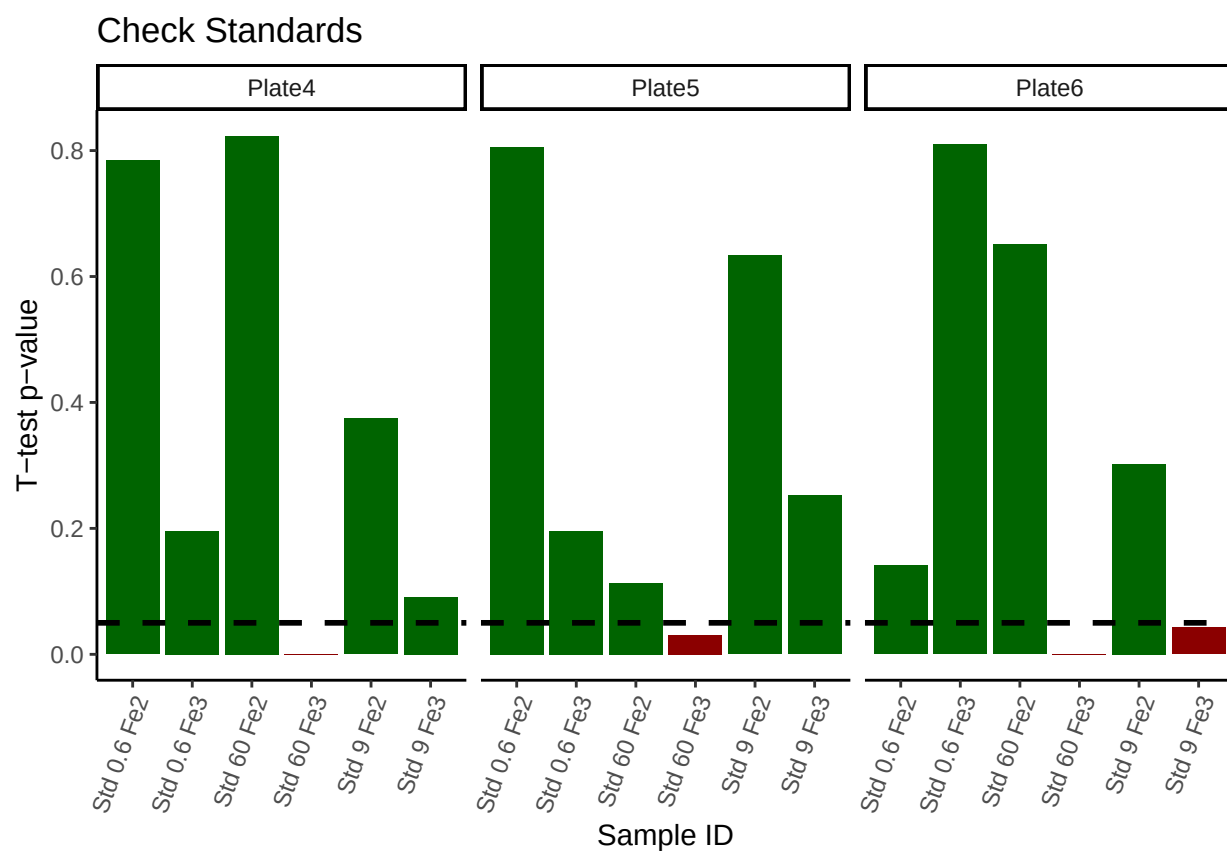
Check standards QAQC

```
## [1] "Plate4"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate5"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
```

```

## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate6"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"

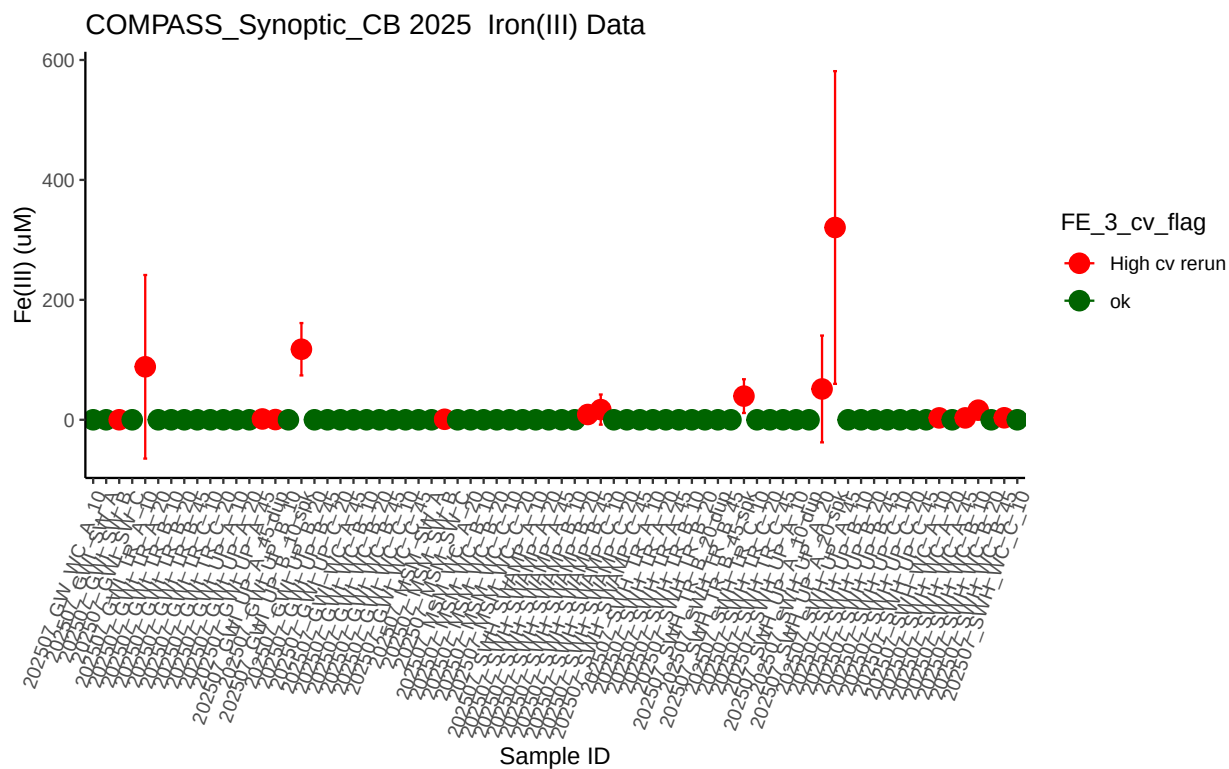
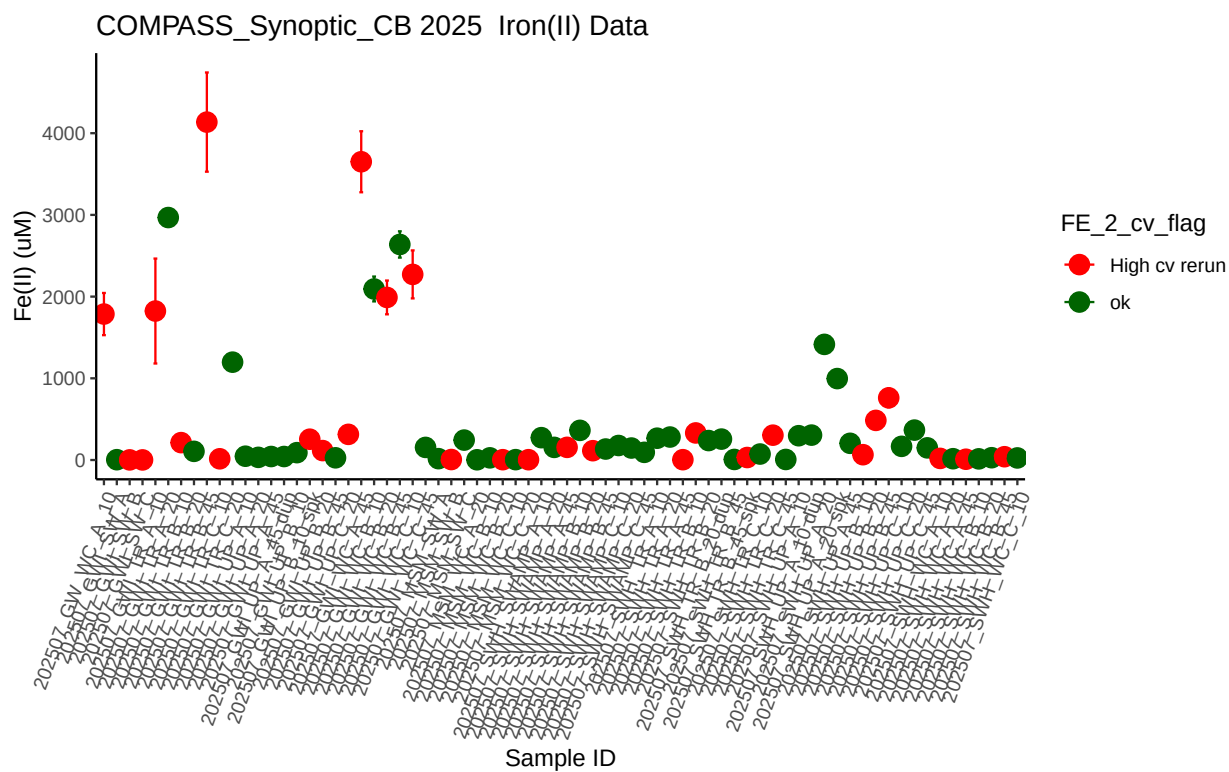
```



#No Matrix Checks We do not use matrix checks for the Fe samples because our matrix checks are not made with Trace metal grade reagents. If we want to do this in the future, we need to use trace metal grade sodium chloride and sodium bicarbonate for making the matrix checks

Subset Data and Calculate Concentrations

Calculate Averages across wells, std. dev, and cv.



Remove Bad Reps

Flagged data

Table 3: High CV Samples

ID full	FE 2 mean	FE 2 cv	FE 3 mean	FE 3 cv	FE 2 cv flag	FE 3 cv flag
Plate4_1_202507_MSM_SW_B	5.401513	4.561592	1.305246	76.90282	ok	High cv rerun
Plate4_8_202507_GWI_UP_A_45	42.747494	9.961267	2.048511	62.48397	ok	High cv rerun
Plate4_8_202507_GWI_UP_B_10 spk	234.165511	5.050684	138.380707	25.35083	ok	High cv rerun
Plate5_8_202507_SWH_UP_A_20 spk	996.579379	8.581680	458.133277	32.77132	ok	High cv rerun
Plate6_1_202507_SWH_SWAMP_B_45	134.386170	3.987870	25.509775	112.87457	ok	High cv rerun
Plate6_1_202507_SWH_TR_B_45 spk	32.755240	6.017846	53.462137	37.84262	ok	High cv rerun
Plate6_1_202507_SWH_WC_B_45	37.285157	6.608393	2.634419	25.76736	ok	High cv rerun
Plate4_8_202507_GWI_TR_B_10	231.377870	10.223069	0.000000	0.00000	High cv rerun	ok

Table 4: Samples Above the Detection Limit that should be rerun at a higher dilution

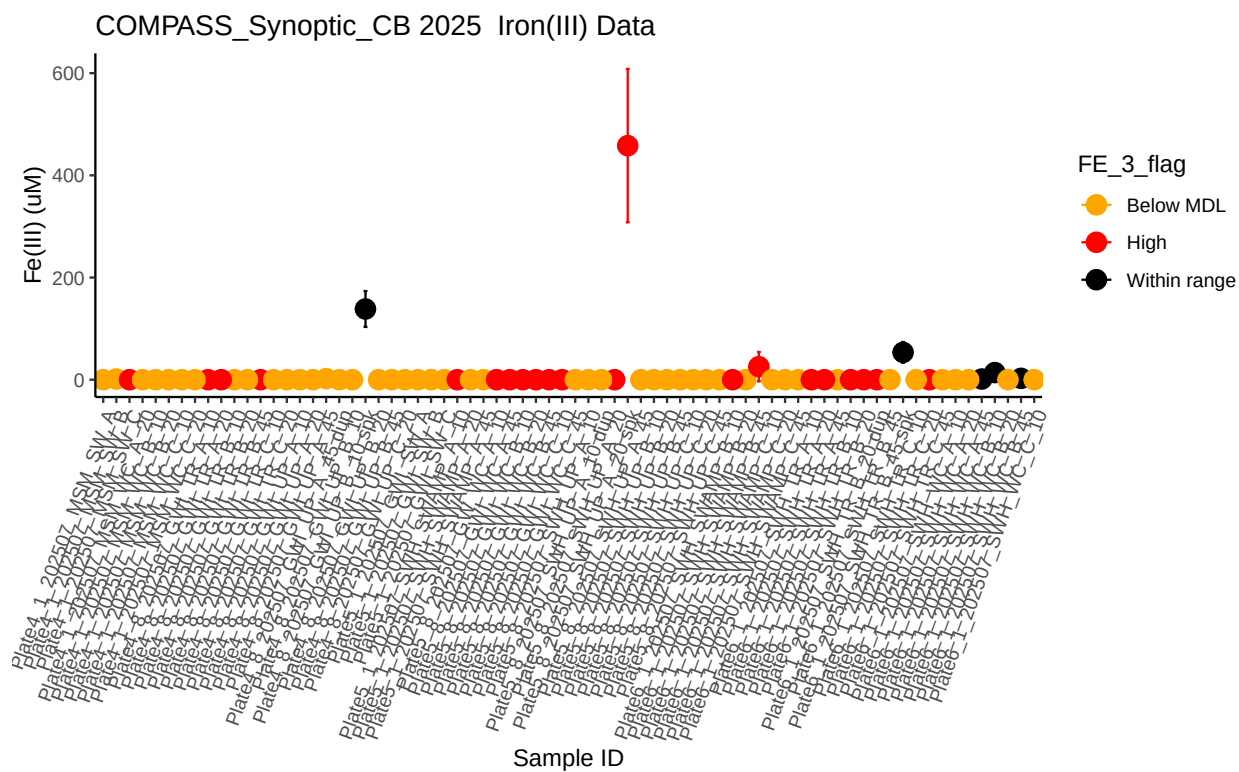
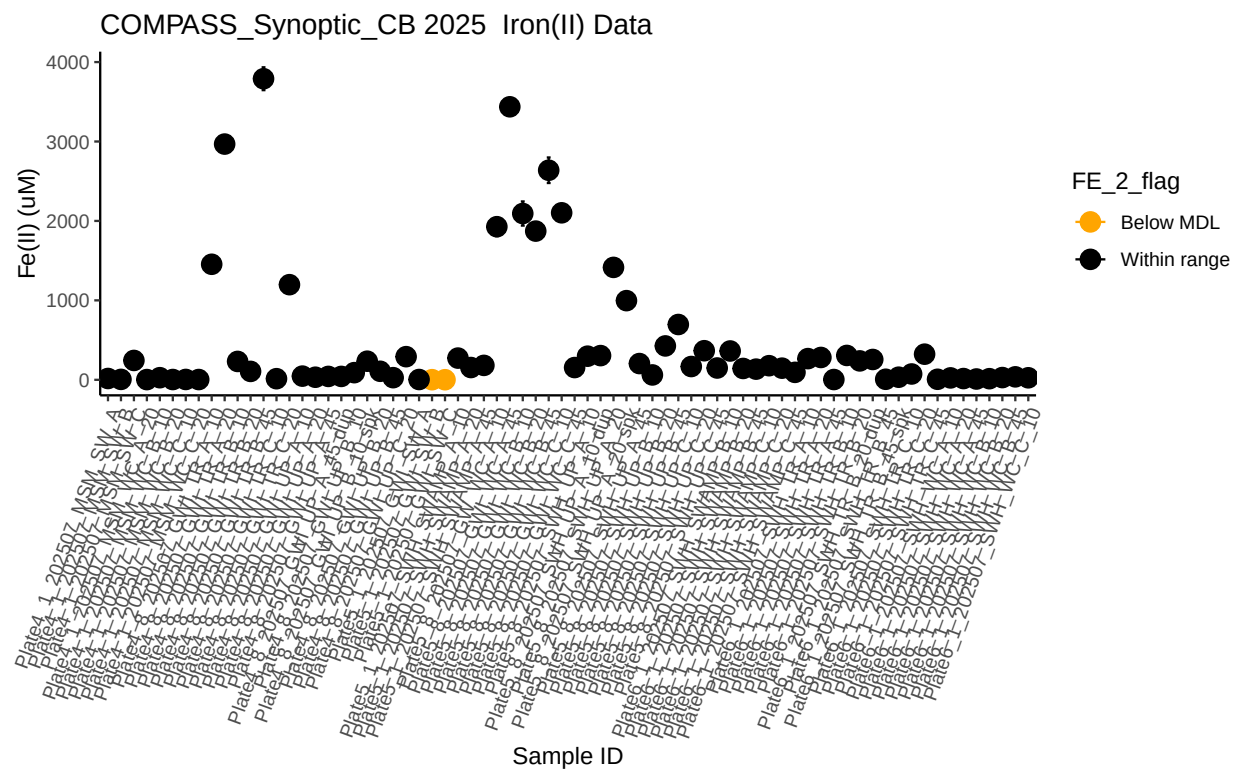
ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate4_8_202507_GWI_TR_A_10	1454.5543	0.00000	1454.5543	Within range	High
Plate5_8_202507_SWH_UP_A_20	1415.2993	0.00000	1415.2993	Within range	High
Plate6_1_202507_SWH_SWAMP_B_45	134.3862	25.50977	159.8959	Within range	High
Plate4_8_202507_GWI_TR_B_45	3790.2264	0.00000	3790.2264	Within range	High
Plate5_8_202507_GWI_WC_A_45	3436.9314	0.00000	3436.9314	Within range	High
Plate5_8_202507_GWI_WC_B_20	1872.0847	0.00000	1872.0847	Within range	High
Plate5_8_202507_GWI_WC_C_10	2102.2617	0.00000	2102.2617	Within range	High
Plate5_8_202507_GW_WC_A_10	1927.3985	0.00000	1927.3985	Within range	High
Plate6_1_202507_SWH_TR_B_10	306.7215	0.00000	306.7215	Within range	High
Plate6_1_202507_SWH_TR_C_20	322.7804	0.00000	322.7804	Within range	High
Plate4_1_202507_MSM_SW_C	243.8243	0.00000	243.8243	Within range	High
Plate4_8_202507_GWI_TR_A_20	2967.6558	0.00000	2967.6558	Within range	High
Plate5_1_202507_SWH_SWAMP_A_10	273.4143	0.00000	273.4143	Within range	High
Plate5_8_202507_GWI_WC_B_10	2093.3401	0.00000	2093.3401	Within range	High
Plate5_8_202507_GWI_WC_B_45	2638.1518	0.00000	2638.1518	Within range	High
Plate6_1_202507_SWH_SWAMP_B_10	362.7789	0.00000	362.7789	Within range	High
Plate6_1_202507_SWH_TR_A_10	266.8718	0.00000	266.8718	Within range	High
Plate6_1_202507_SWH_TR_A_20	281.2950	0.00000	281.2950	Within range	High
Plate6_1_202507_SWH_TR_B_20	238.0253	0.00000	238.0253	Within range	High

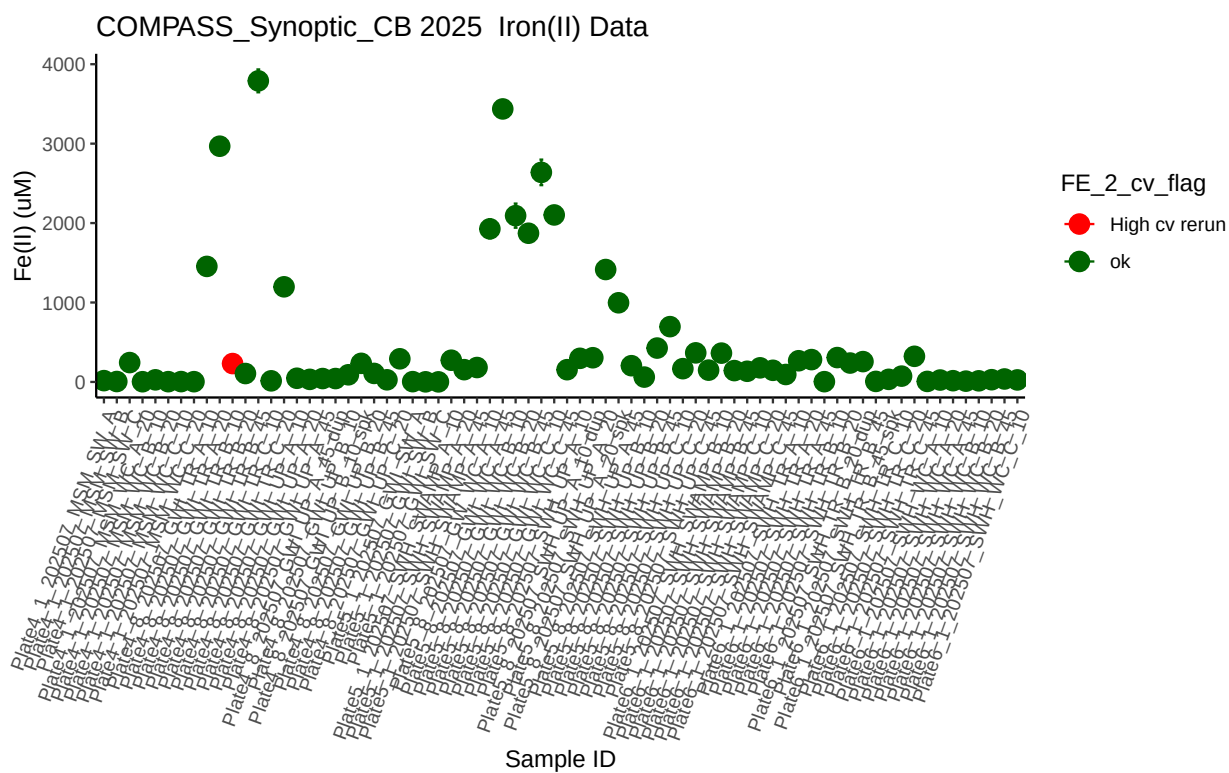
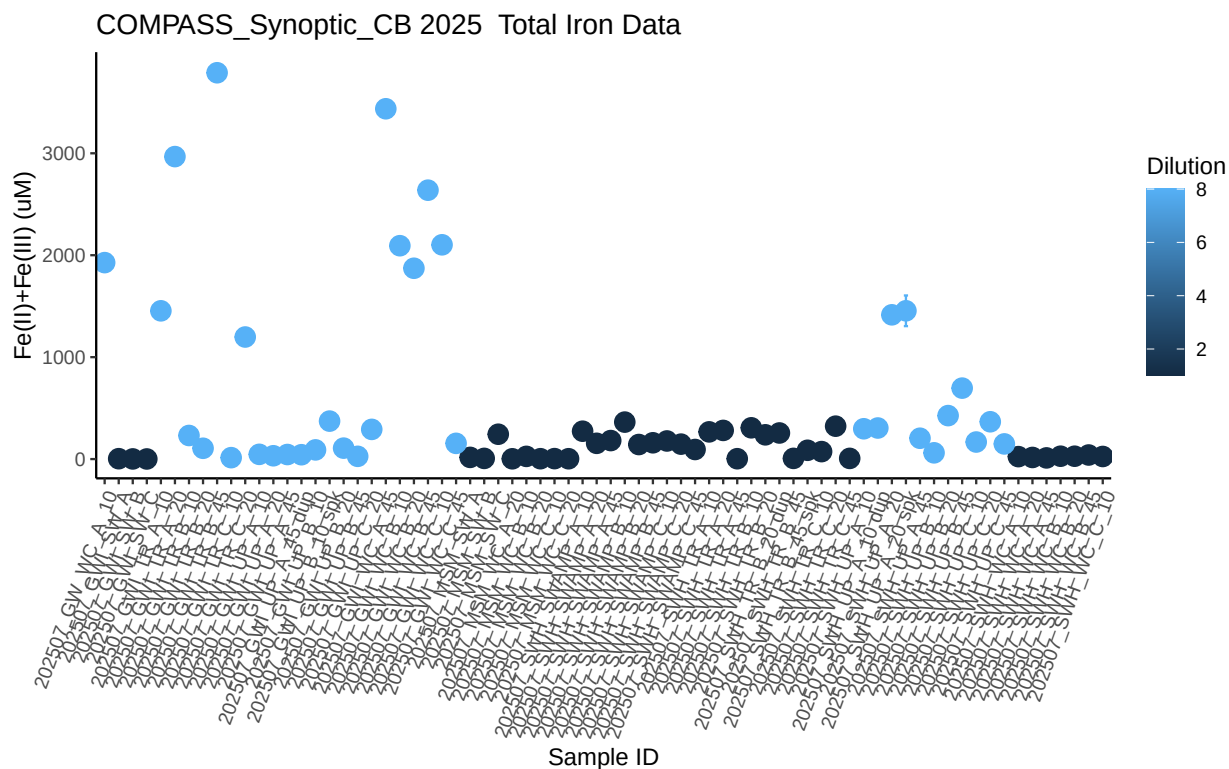
Table 5: Samples Below the Detection Limit that should be rerun at a lower dilution

ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate4_8_202507_GWI_UP_A_45	42.74749	2.048511	44.79601	Within range	Below MDL
Plate4_8_202507_GWI_TR_B_10	231.37787	0.000000	231.37787	Within range	Below MDL
Plate4_8_202507_GWI_TR_C_10	13.94187	0.000000	13.94187	Within range	Below MDL
Plate4_8_202507_GWI_UP_B_20	107.32784	0.000000	107.32784	Within range	Below MDL
Plate4_8_202507_GWI_UP_C_20	291.31215	0.000000	291.31215	Within range	Below MDL
Plate5_8_202507_SWH_UP_B_10	61.33177	0.000000	61.33177	Within range	Below MDL
Plate5_8_202507_SWH_UP_B_20	426.51274	0.000000	426.51274	Within range	Below MDL
Plate5_8_202507_SWH_UP_B_45	696.21921	0.000000	696.21921	Within range	Below MDL
Plate4_8_202507_GWI_TR_B_20	105.93402	0.000000	105.93402	Within range	Below MDL
Plate4_8_202507_GWI_TR_C_20	1197.61252	0.000000	1197.61252	Within range	Below MDL
Plate4_8_202507_GWI_UP_A_10	46.46435	0.000000	46.46435	Within range	Below MDL

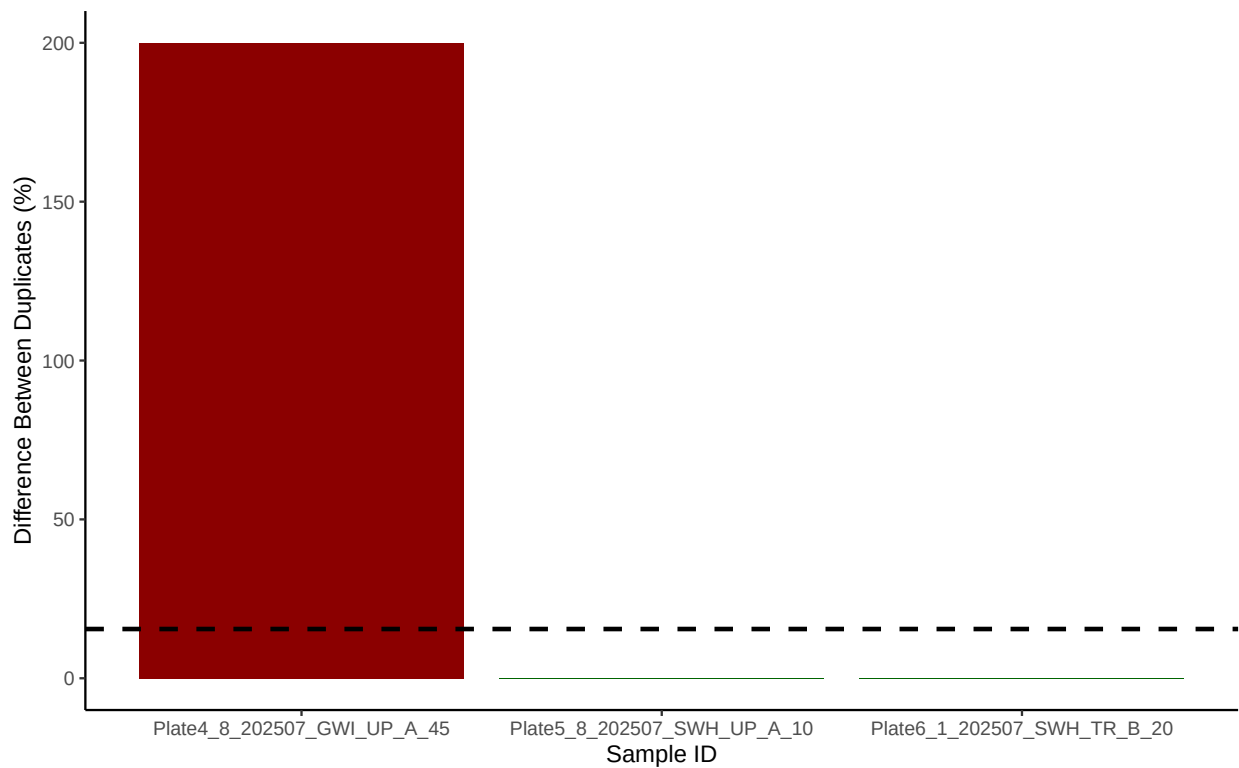
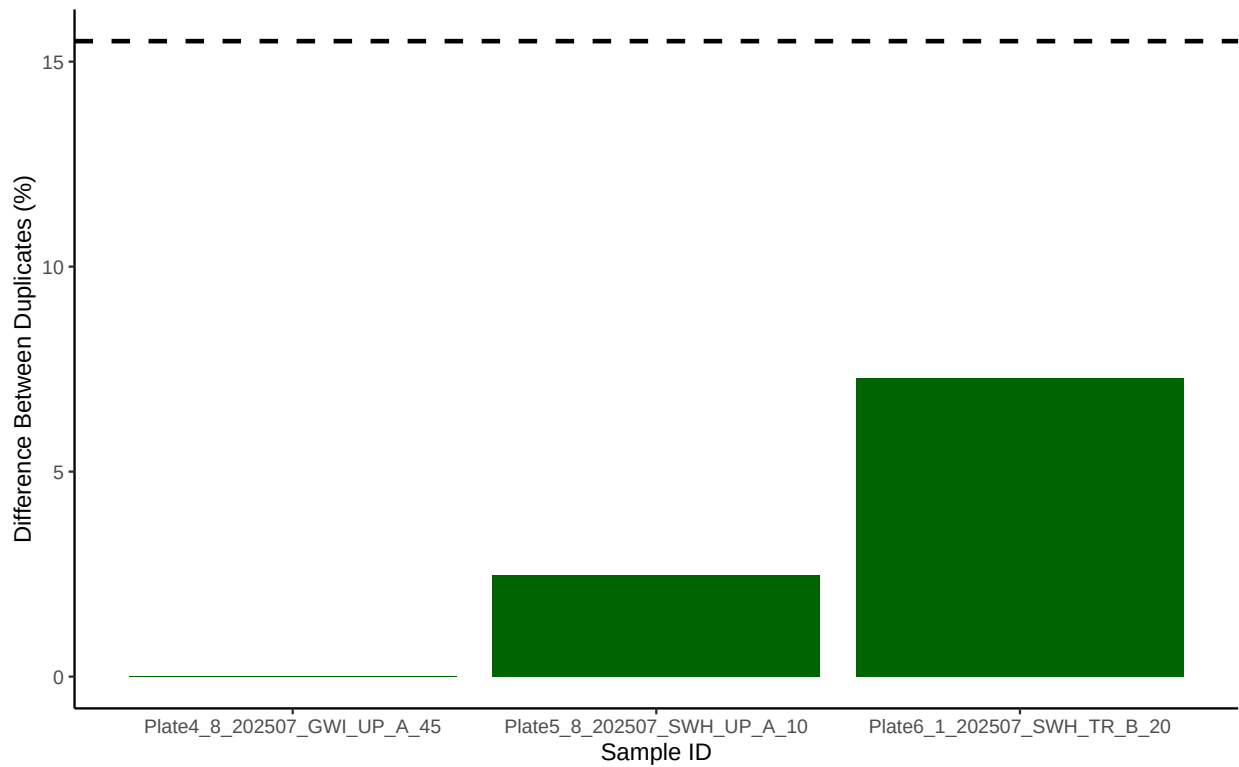
Plate4_8_202507_GWI_UP_A_20	32.52614	0.000000	32.52614	Within range	Below MDL
Plate4_8_202507_GWI_UP_B_10	90.13739	0.000000	90.13739	Within range	Below MDL
Plate4_8_202507_GWI_UP_B_45	26.95086	0.000000	26.95086	Within range	Below MDL
Plate5_8_202507_GWI_WC_C_45	154.25314	0.000000	154.25314	Within range	Below MDL
Plate5_8_202507_SWH_UP_A_10	297.35204	0.000000	297.35204	Within range	Below MDL
Plate5_8_202507_SWH_UP_A_45	204.43067	0.000000	204.43067	Within range	Below MDL
Plate5_8_202507_SWH_UP_C_10	167.26213	0.000000	167.26213	Within range	Below MDL
Plate5_8_202507_SWH_UP_C_20	366.11385	0.000000	366.11385	Within range	Below MDL
Plate5_8_202507_SWH_UP_C_45	148.67785	0.000000	148.67785	Within range	Below MDL

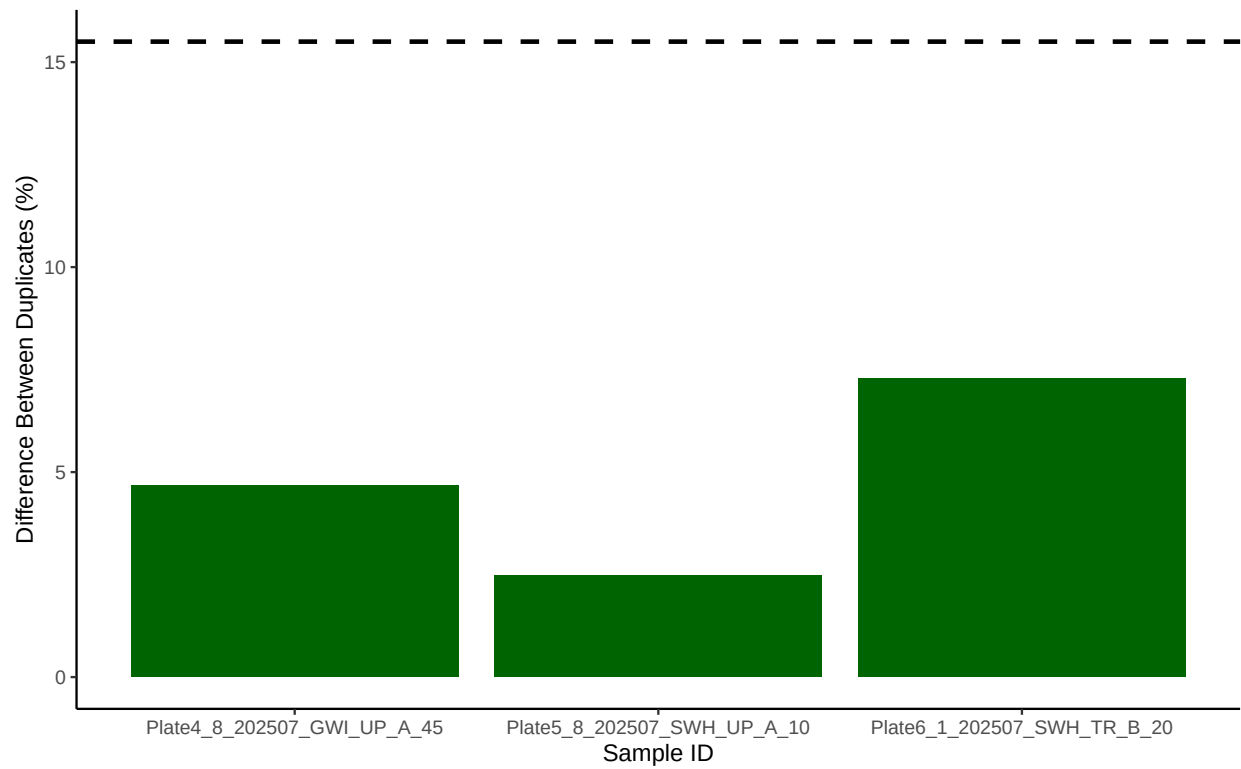
Plot data after bad reps were removed





Check Dups for QAQC

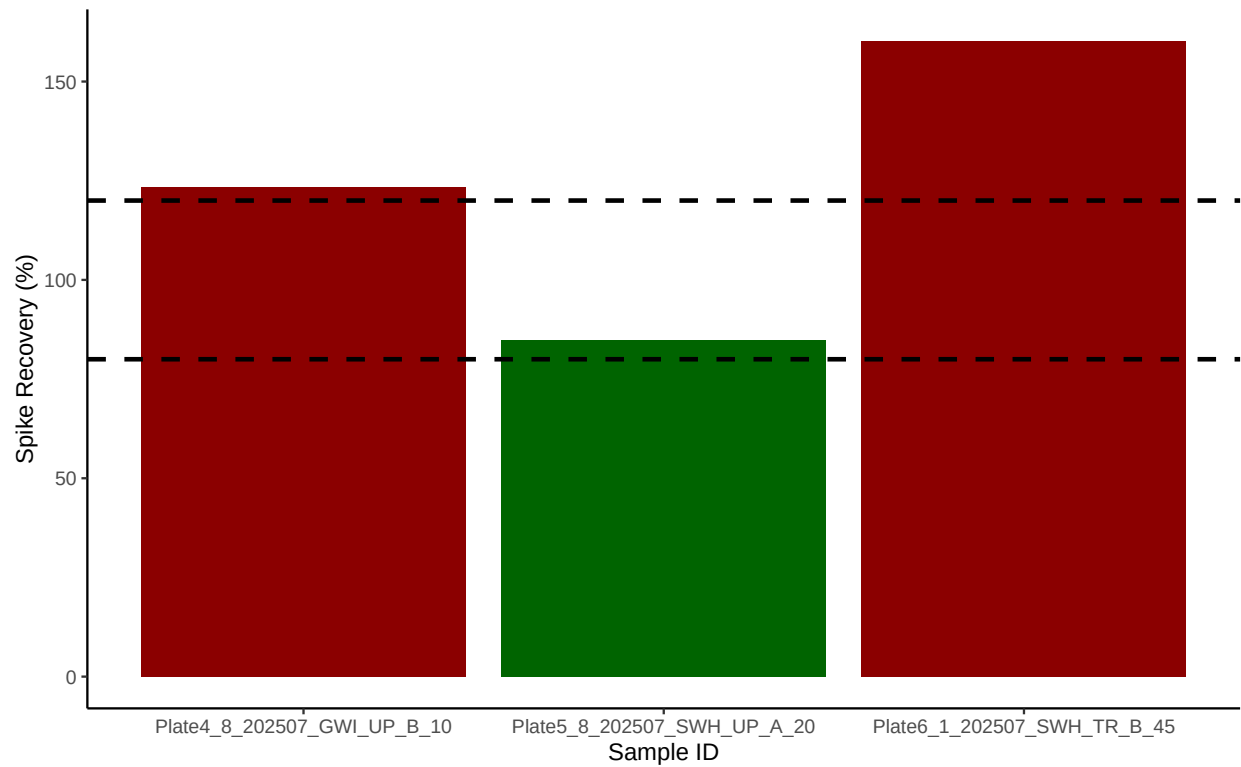




```
## [1] ">60% of Fe 2 Duplicates are within <10%"
```

```
## [1] ">60% of Fe 3 Duplicates are within <10%"
```


Check the spks for QAQC



```
## [1] "<60% of Spikes are out of range - REASSESS"
```

Export full data then just final data